

Similarity and Distance Measures

- ❑ Data Matrix versus Dissimilarity Matrix
- ❑ Proximity Measures for Nominal Attributes
- ❑ Proximity Measures for Binary Attributes
- ❑ Dissimilarity of Numeric Data: Minkowski Distance
- ❑ Proximity Measures for Ordinal Attributes
- ❑ Dissimilarity for Attributes of Mixed Types
- ❑ Cosine Similarity
- ❑ Measuring Similar Distributions: The Kullback-Leibler Divergence
- ❑ Capturing Hidden Semantics in Similarity Measures

Similarity, Dissimilarity, and Proximity

□ **Similarity measure or similarity function**

- A real-valued function that quantifies the similarity between two objects
- Measure how two data objects are alike: The higher value, the more alike
- Often falls in the range $[0,1]$: 0: no similarity; 1: completely similar

□ **Dissimilarity (or distance) measure**

- Numerical measure of how different two data objects are
- In some sense, the inverse of similarity: The lower, the more alike
- Minimum dissimilarity is often 0 (i.e., completely similar)
- Range $[0, 1]$ or $[0, \infty)$, depending on the definition

□ **Proximity** usually refers to either similarity or dissimilarity